

AMAR DHILLON

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Education

University of Maryland, College Park

B.S. in Computer Science, Minor in Statistics; GPA: 3.5

May 2028

College Park, MD

Projects

IEX Order Book & Matching Engine | C++17, STL, POSIX, Linux perf

Jul 2026 – Aug 2026

- Built a C++17 order book and matching engine that replayed **50.9M** real IEX DEEP+ messages with no resync, matching 99,122 of 99,123 fills exactly and 100% of top-of-book prices
- Indexed orders in a hash map holding iterators into per-price FIFO queues (std::map levels, O(1) cancels); a separate 2M-event profiling run put the per-update p99 tail at 1.2 μ s
- Sustained **1.26M events/sec** end-to-end over the full replay; the matching engine alone averages 686.7 ns/event (1.46M/s) with the CPU fixed at its 1.40 GHz base clock
- Traced a $1.65\times$ phantom regression to a CPU-governor artifact using perf counters and a null control book; core pinning and a fixed clock cut run-to-run variance from $\sim 6\%$ to 0.06%
- Benchmarked B+ tree vs LSM stores for the replay's event log; the B+ tree gave $12.8\times$ read throughput for 27% more disk; documented adaptive log batch sizing as a negative result

Experience

Amazon Web Services

Sep 2026 – Dec 2026

Software Development Engineer Intern

Redmond, WA

- Writing Java policies that tune upper/lower bounds on EC2 capacity reserved for Amazon Redshift operations like patching, balancing availability against cost; initial backtests complete

Capital One

Jun 2026 – Aug 2026

Software Engineering Intern

Richmond, VA

- Precomputed a category-keyed tree index of popular workflows from a 270K+ row PostgreSQL table (Python/SQLAlchemy, Redis cache) to minimize DB queries under access limits
- Cut LLM workflow generation time **65%** to under 15s by splitting a Claude agent skill into sub-skills backed by Python search/validation tools, guided by CloudWatch + OpenSearch latency logs

UMD App Development Contracting (Client: Amazon Leo)

Jan 2026 – Present

Software Engineer (Contract)

College Park, MD

- Reduced end-to-end latency **34%** for a pipeline routing 6,000+ scraped research documents to inference by replacing S3 staging hops with Kinesis streams feeding Lambda

UMD Institute for Advanced Computer Studies (UMIACS)

Dec 2025 – Present

Machine Learning Research Intern

College Park, MD

- Built a Python pipeline for PhysTwin (PhD-led digital twin, targeting ICLR) turning AnyGrasp output on multi-camera images into Open3D overlays for a VLM; tuned its prompts to pick correct grasps and judge depth

TerpLabs

Jun 2025 – Present

Co-Founder, Software Engineer

College Park, MD

- Co-founded a 110+ member campus product studio that has shipped 5 live products, including Tortuga, a class scheduler with **20,000+ visits** in its first 30 days; built Redis caching for its Prisma ORM queries

Technical Skills

Languages: C++17, C, Python, Java, SQL, TypeScript

Systems & Performance: Linux perf (hardware counters, top-down analysis), POSIX, STL, B+ trees, LSM trees

Cloud & Tools: AWS (Lambda, Kinesis, Fargate, CDK, CloudWatch), Redis, PostgreSQL, Docker, GitHub Actions, Pytest

Certification: AWS Certified Developer – Associate (Nov 2024 – Nov 2027)